

Yixiao Peng

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New Brunswick, NJ

EDUCATION

Rutgers University

New Brunswick, NJ

*Master of Business & Science (MBS), Computer and Information
Sciences (UXD Concentration)*

Sep 2025-May 2027 (*Expected*)

Fuzhou University, Xiamen Academy of Arts and Design

Xiamen, China

Bachelor of Engineering in Industrial Design

Sep 2021-Jun 2025

EXPERIENCE

AI-Driven User Experience Design

Mar 2024-May 2024

Advisor: Prof. Xiang Chen, University of California, Los Angeles

- Conducted 8+ user interviews, surveys, and usability testing to identify user needs, pain points, and usability barriers in AI-assisted image description workflows
- Synthesized qualitative and survey data to uncover patterns in user behavior, informing key design decisions and feature prioritization
- Evaluated how users interacted with and refined AI-generated descriptions, revealing opportunities to improve clarity, control, and user trust
- Translated research insights into design improvements for an interactive web platform, enabling users to iteratively refine AI-generated outputs

Simulated Lunar Illumination and Human Perception

Oct 2024-Dec 2024

- Investigated how variations in brightness and color temperature influenced human perception and object recognition in low-light environments
- Analyzed perceptual responses across simulated lighting scenarios to better understand visual experience and cognitive processing
- Developed and validated a perception model to better understand user visual experience under varying lighting conditions
- Applied research insights to inform human-centered design considerations related to visibility and environmental interaction

Development and Design of Dehua Cultural and Creative Products

Mar 2023-Jun 2024

- Conducted literature reviews, field research, and market analysis to understand user preferences and consumer behavior in the Dehua porcelain market
- Segmented users based on demographics and purchasing patterns to identify target groups and inform product positioning
- Translated research findings into culturally informed product concepts aligned with user needs and market trends
- Collaborated with stakeholders and external partners to support product development and market strategy execution

EXPERTISE

Research Methods:

User Interviews · Surveys · Usability Testing · Contextual Inquiry · Research Synthesis · Qualitative Analysis · Persona Development · Journey Mapping

Research & Analysis:

User Research · Human-Computer Interaction (HCI) · Insight Generation · Behavioral Analysis · Problem Framing

Tools:

Figma · FigJam · Excel · Miro · Google Forms · Blender · Rhino · Tableau

Selected Publications

- Dai, S., Peng, Y., Chen, K., & Lyu, H. (2025). CapsoAn LLM-based Verbal Aphasia Assistance App. *Applied and Computational Engineering*, 131(1), 106–110. <https://doi.org/10.54254/2755-2721/2024.20570>
- Ke, Y., Peng, Y., Chen, Y., Cai, Z., Jiang, A., Foing, B., & Yang, H. (2025). The Impact of Simulated Lunar Dynamic Lighting Conditions on Object Information Recognition and Visual Perception. In *Proceedings of*

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HCI International 2025 (LNCS, Springer)